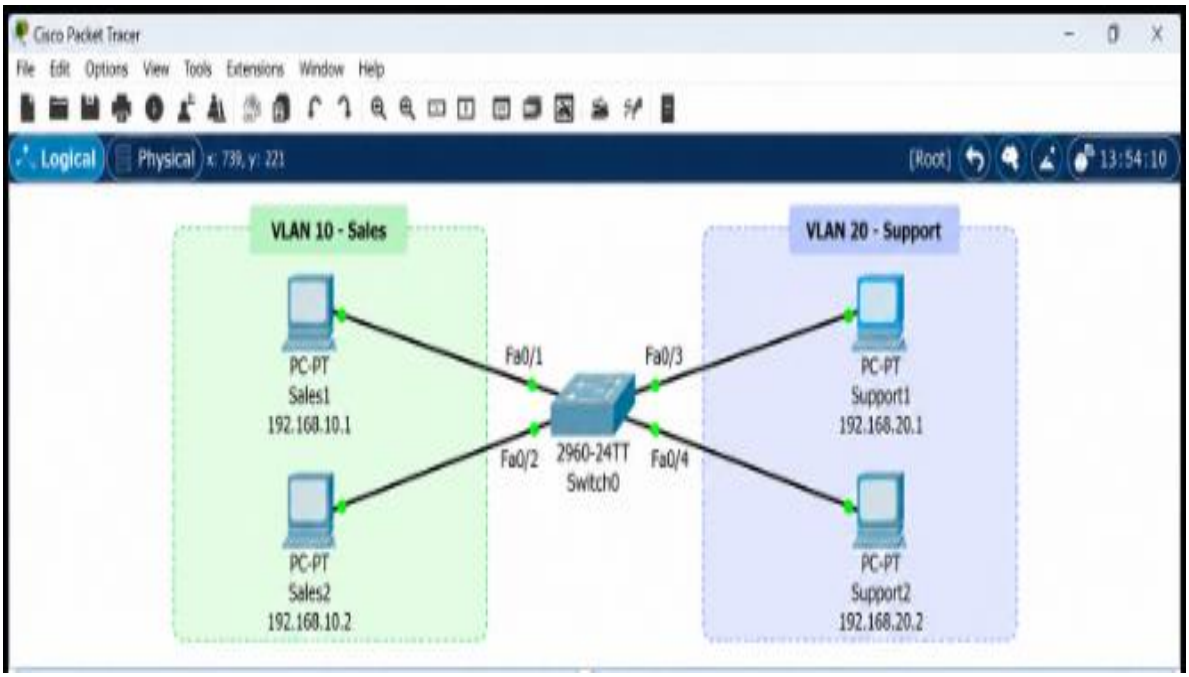
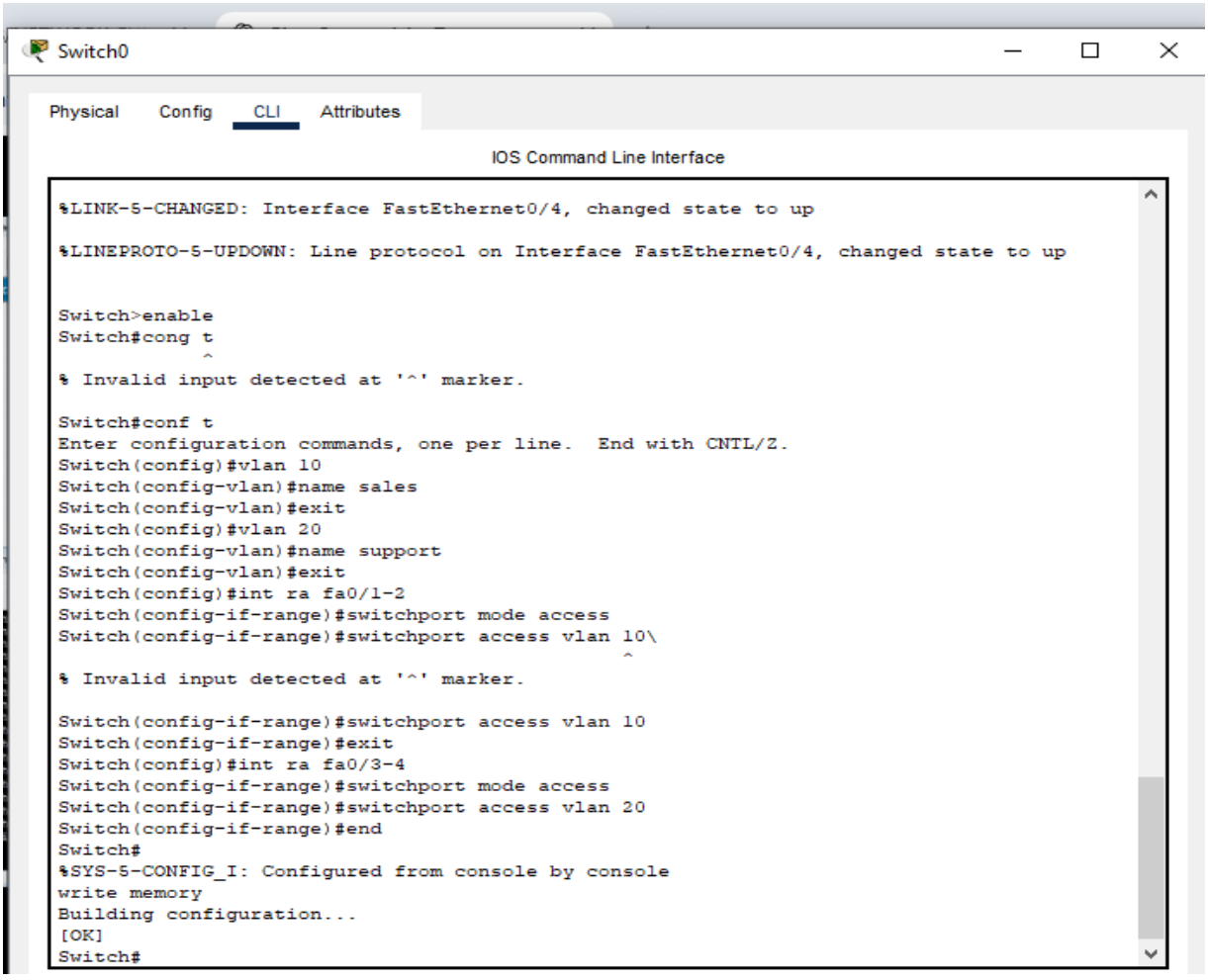
1. Log in to Cisco Packet Tracer or any network simulation tool, add a switch, and create two VLANs named 'Sales' and 'Support' with VLAN IDs 10 and 20.



Click on the switch and open CLI tab and type the following command shown in the image given below



```
Switch#
Switch#sh vlan brief

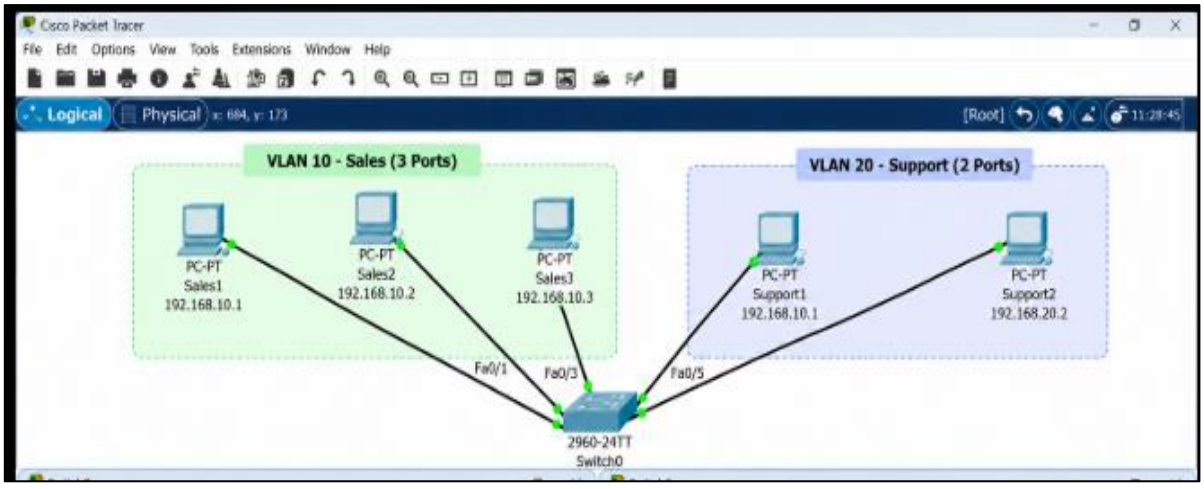
VLAN Name                              Status    Ports
---- --------------------------------- --------- -------------------------------
1    default                           active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                 Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                 Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                 Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                 Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                 Gig0/1, Gig0/2
10   sales                             active    Fa0/1, Fa0/2
20   support                           active    Fa0/3, Fa0/4
1002 fddi-default                      active
1003 token-ring-default                active
1004 fddinet-default                   active
1005 trnet-default                     active
Switch#
```

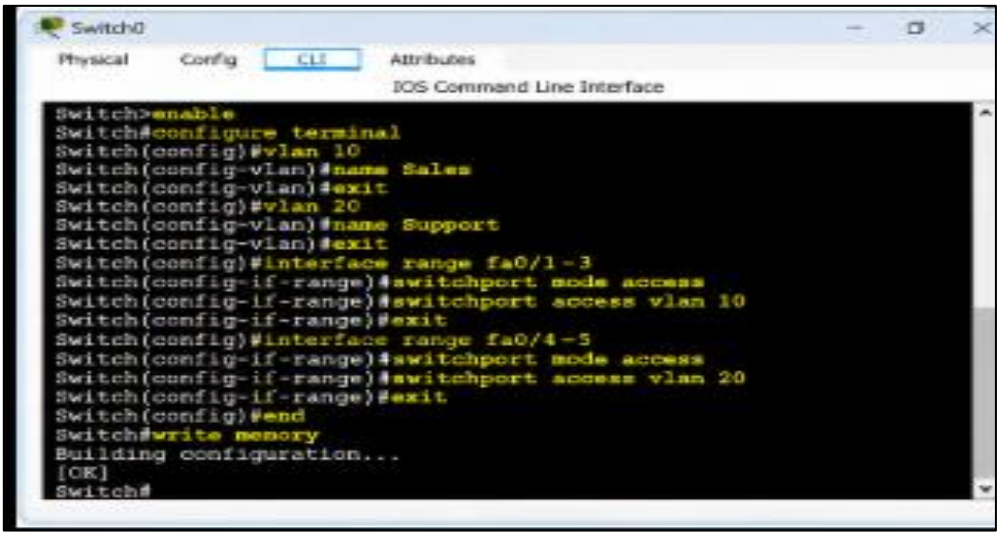
Copy Paste

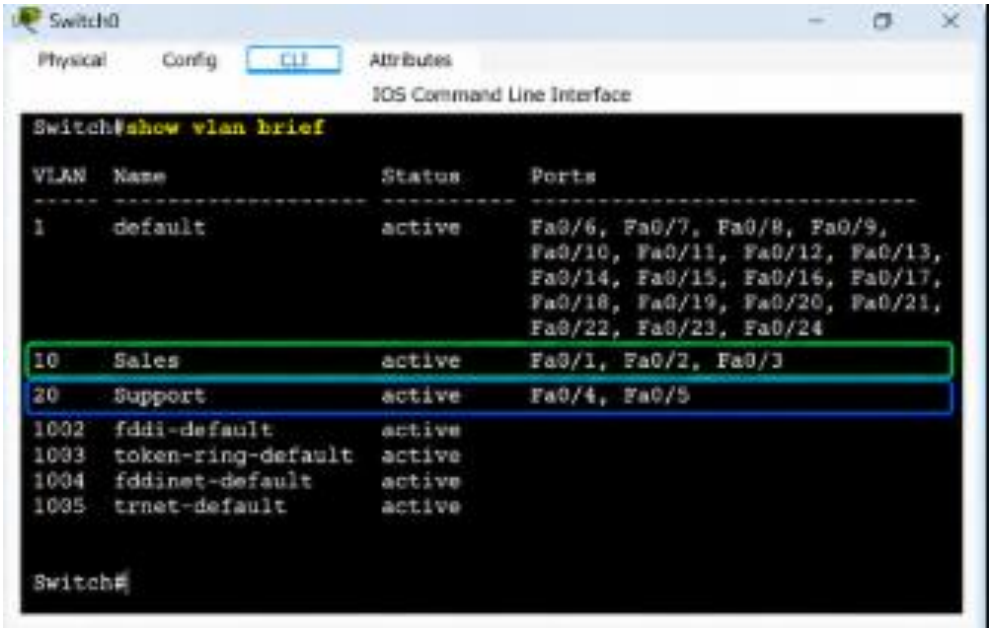
Vlan 10 and vlan 20 with name sales and support is visible in the above image.

2. Assign three different switch ports to VLAN 10 (Sales) and two ports to VLAN 20 (Support), then verify the VLAN membership using the show vlan command.

Hint: Use the interface range command to select multiple ports efficiently.

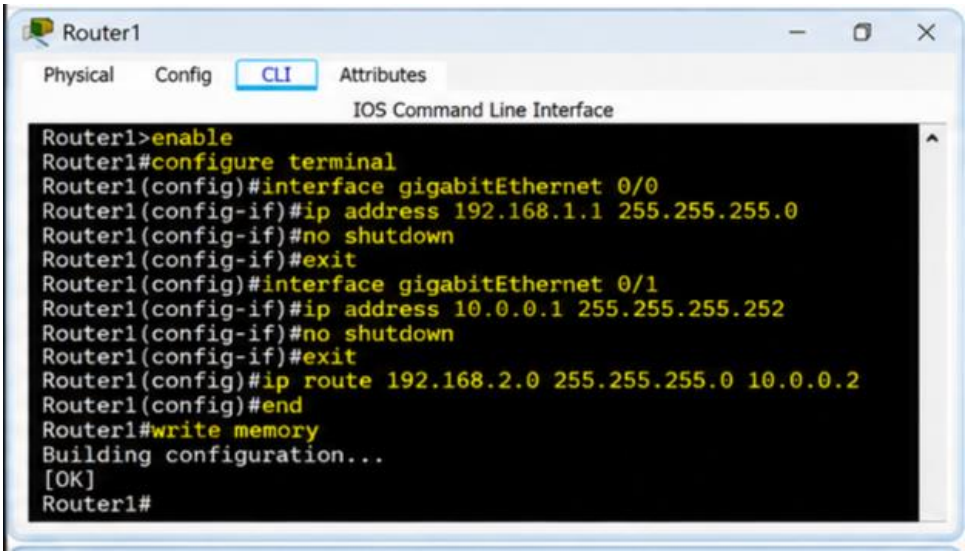


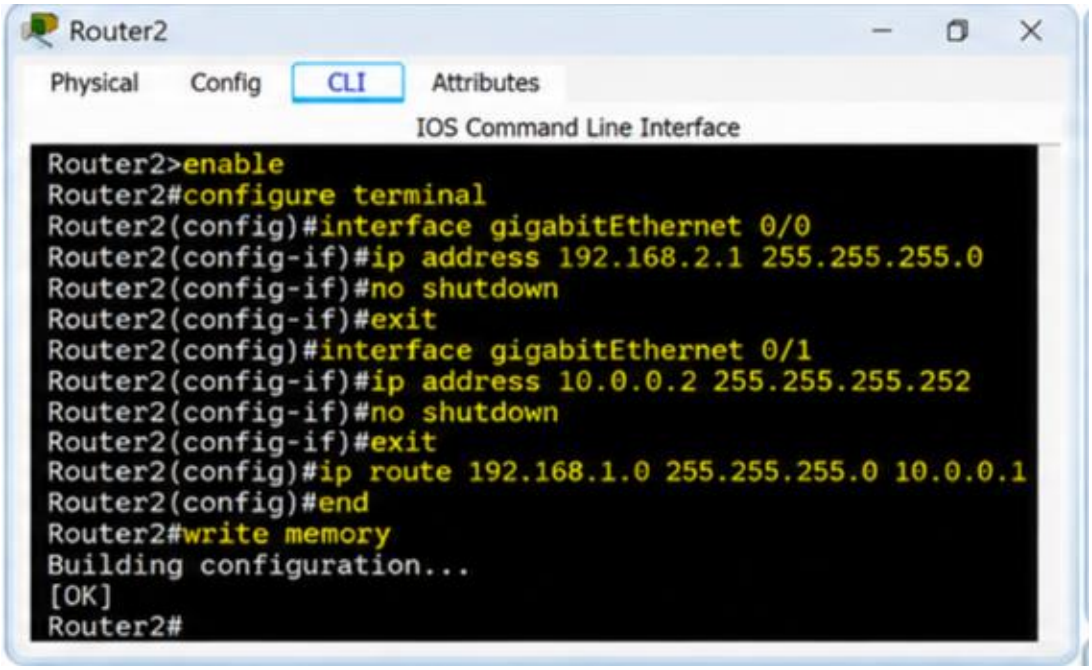


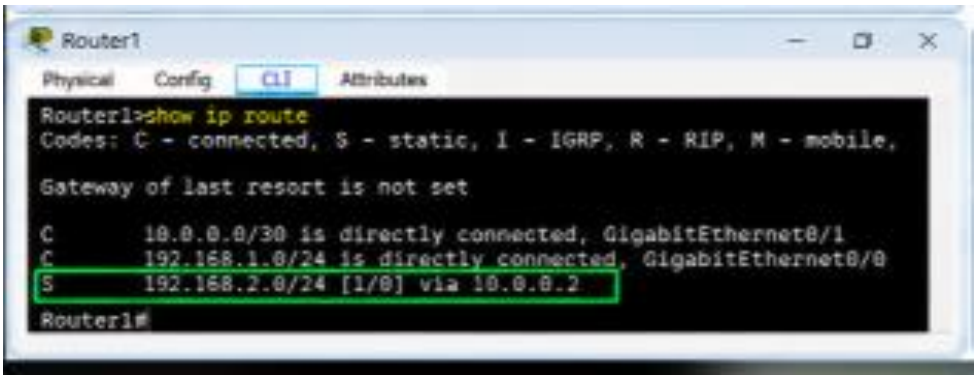


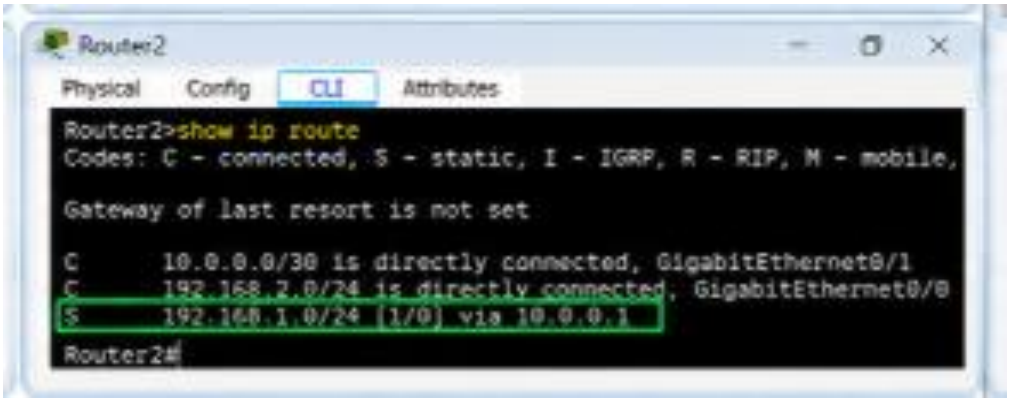
3. Connect two routers using a serial or Ethernet link in your simulation tool, assign IP addresses to their interfaces, and configure a static route on Router 1 to reach the LAN behind Router 2.

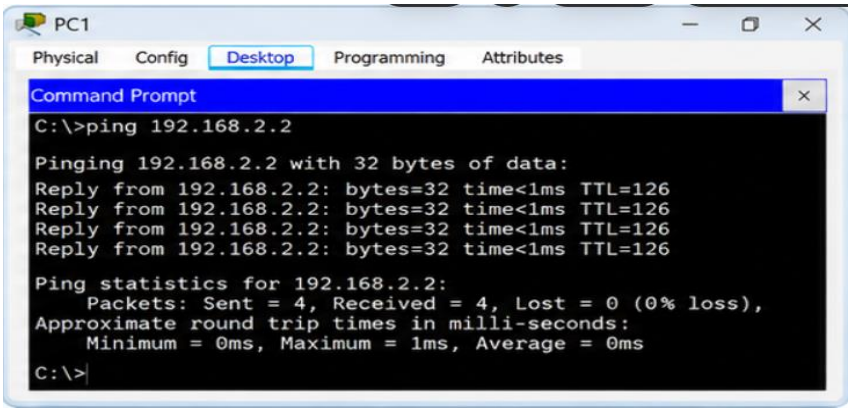


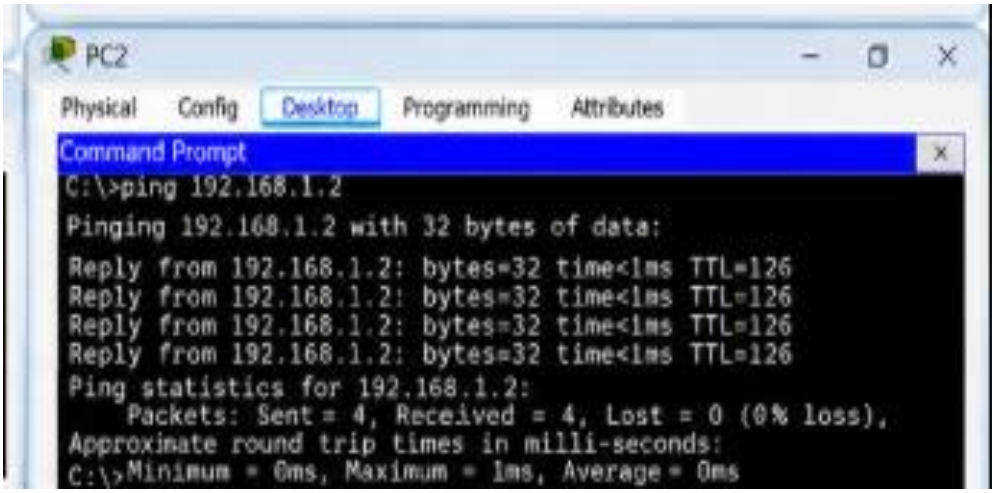



Router2
Physical
Config
CLI
Attributes
IOS Command Line Interface
Router2>enable
Router2#configure terminal
Router2(config)#interface gigabitEthernet 0/0
Router2(config-if)#ip address 192.168.2.1 255.255.255.0
Router2(config-if)#no shutdown
Router2(config-if)#exit
Router2(config)#interface gigabitEthernet 0/1
Router2(config-if)#ip address 10.0.0.2 255.255.255.252
Router2(config-if)#no shutdown
Router2(config-if)#exit
Router2(config)#ip route 192.168.1.0 255.255.255.0 10.0.0.1
Router2(config)#end
Router2#write memory
Building configuration...
[OK]
Router2#


Router1
Physical
Config
CLI
Attributes
Router1>show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile,
Gateway of last resort is not set
C 10.0.0.0/30 is directly connected, GigabitEthernet0/1
C 192.168.1.0/24 is directly connected, GigabitEthernet0/0
S 192.168.2.0/24 [1/0] via 10.0.0.2
Router1#

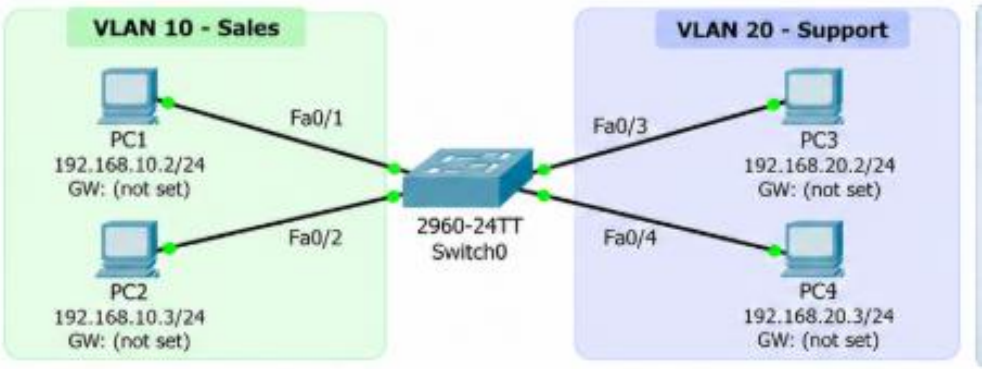

Router2
Physical
Config
CLI
Attributes
Router2>show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile,
Gateway of last resort is not set
C 10.0.0.0/30 is directly connected, GigabitEthernet0/1
C 192.168.2.0/24 is directly connected, GigabitEthernet0/0
S 192.168.1.0/24 [1/0] via 10.0.0.1
Router2#

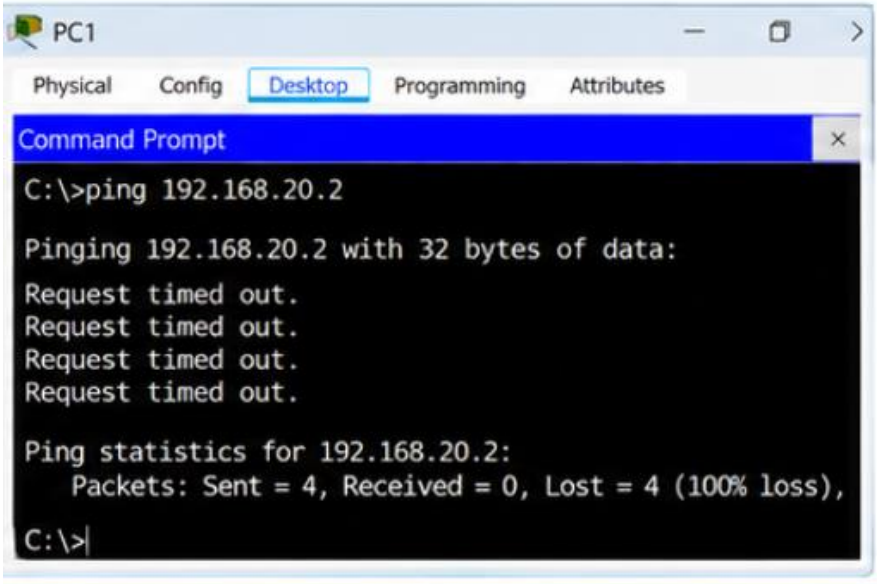

PC1
Physical
Config
Desktop
Programming
Attributes
Command Prompt
C:\>ping 192.168.2.2
Pinging 192.168.2.2 with 32 bytes of data:
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Ping statistics for 192.168.2.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>

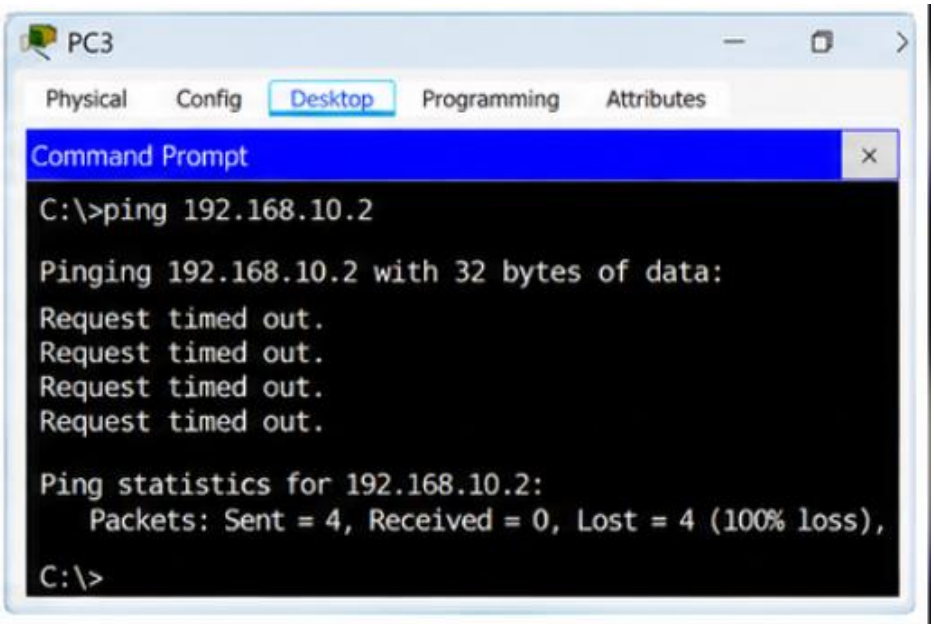


4. Simulate a scenario where a device in the 'Sales' VLAN tries to ping a device in the 'Support' VLAN and explain why the ping fails, then suggest what configuration would allow inter-VLAN communication.

Hint: Think about router-on-a-stick or Layer 3 switch concepts.

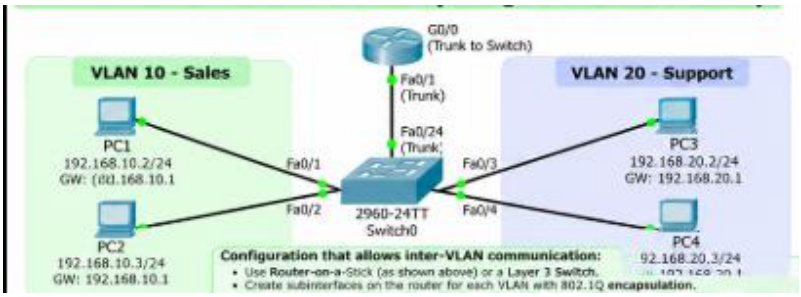


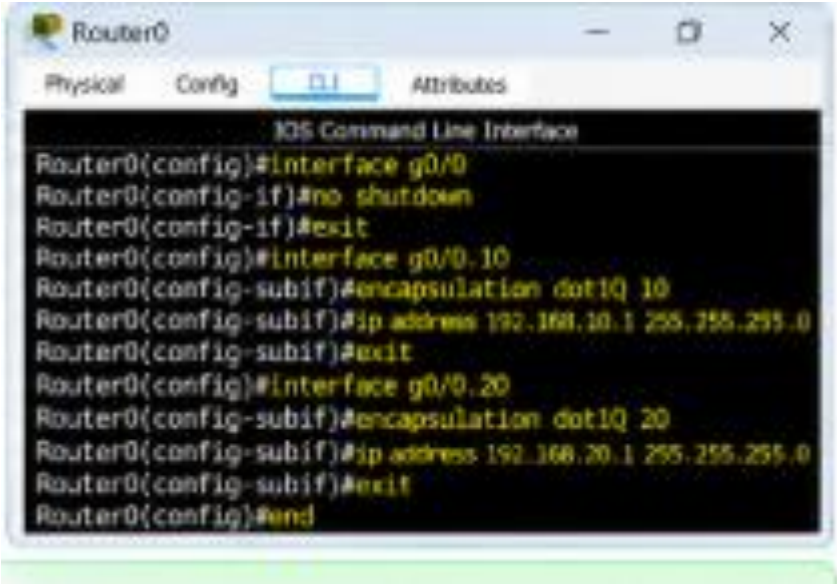


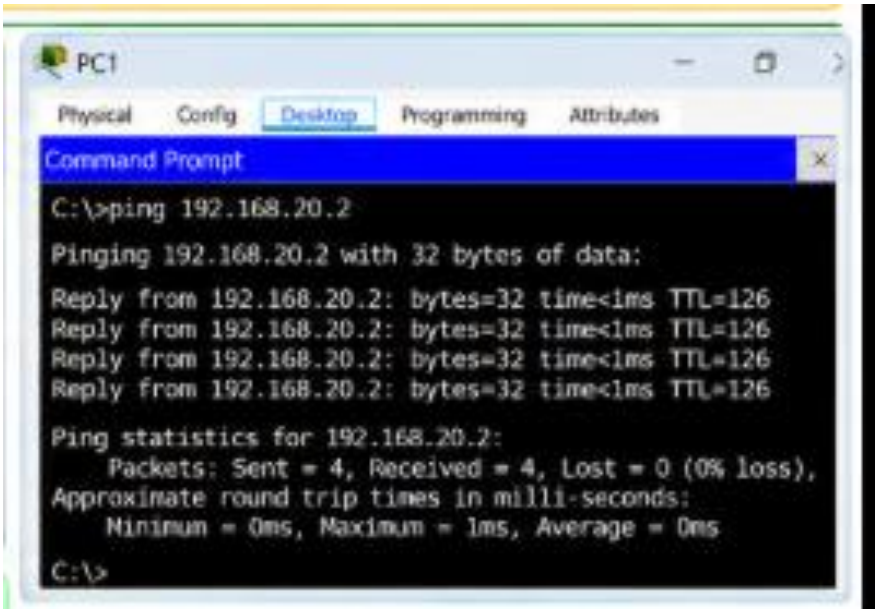


Why the ping fails?

- VLAN10 (sales) and VLA20 (support) are separate bvroadcast domains
- A layer 2 switch does not allow communication between different vlan by default.
- Each device can communicate only with devices in the same VLAN
- That's why PC1 cannot ping PC3






PC1
Physical
Config
Desktop
Programming
Attributes
Command Prompt
C:\>ping 192.168.20.2
Pinging 192.168.20.2 with 32 bytes of data:
Reply from 192.168.20.2: bytes=32 time<1ms TTL=126
Reply from 192.168.20.2: bytes=32 time<1ms TTL=126
Reply from 192.168.20.2: bytes=32 time<1ms TTL=126
Reply from 192.168.20.2: bytes=32 time<1ms TTL=126
Ping statistics for 192.168.20.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>